15. Which of these rational numbers which have additive inverse?

(A) 7 (B) $\frac{-5}{7}$ (C) 0 (D) all of these

Note

Why $\frac{a}{b}+\frac{c}{d} \neq \frac{a+c}{b+d}$?

A mathematical statement is true only if it is 100% true without any exception. If we do the addition, say $\frac{3}{4}+\frac{1}{4}$ as $\frac{3+1}{4+4}=\frac{4}{8}=\frac{1}{2}$. This is wrong, since $\frac{3}{4}+\frac{1}{4}=\frac{3+1}{4}=\frac{4}{4}=1$.

1.5 Properties of Rational Numbers

Some properties listed here below will be of good use in solving problems.

1.5.1 Closure property/law for the collection $\mathbb{Q}$ of rational numbers

i) Closure property for Addition

For any two rational numbers ***a*** and ***b***, the sum ***a + b*** is also a rational number.

ii) Closure property for Multiplication

For any two rational numbers ***a*** and ***b***, the product ***ab*** is also a rational number.

Illustration

Take $a=\frac{3}{4}$ and $b=\frac{-1}{2}$

Now, $a+b=\frac{3}{4}+\frac{-1}{2}=\frac{3}{4}+\frac{-2}{4}=\frac{3-2}{4}=\frac{1}{4}$ is in $\mathbb{Q}$

Also, $a\times b=\frac{3}{4}\times\frac{-1}{2}=\frac{-3}{8}$ is in $\mathbb{Q}$

Try this

The closure property on integers holds for subtraction and not for division. What about rational numbers? Verify.

1.5.2 Commutative property/law for the collection $\mathbb{Q}$ of rational numbers

i) Commutative property for Addition

For any two rational numbers ***a*** and ***b***, ***a + b = b + a***.

ii) Commutative property for Multiplication

For any two rational numbers ***a*** and ***b***, ***ab = ba*** (ab means $a\times b$ and ba means $b\times a$).

Illustration

Take $a=\frac{-7}{8}$ and $b=\frac{3}{5}$

Now, $a+b=\frac{-7}{8}+\frac{3}{5}=\frac{-7\times5+3\times8}{40}=\frac{-35+24}{40}=\frac{-11}{40}$

Also, $b+a=\frac{3}{5}+\frac{-7}{8}=\frac{3\times8+-7\times5}{40}=\frac{24-35}{40}=\frac{-11}{40}$

Here, we find that $a+b=b+a$ and hence addition is commutative.

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